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NUT, FISHING ROD REEL SEAT, AND FISHING ROD

Abstract

A nut according to an embodiment of the present invention comprises: a nut base portion which has an outer peripheral surface provided with a recessed portion and an inner peripheral surface provided with a female screw portion for movement that is screwed into the male screw portion of the reel seat main body, and of which an end portion is locked to the hood main body of the movable hood; and a cylindrical portion that has a protrusion portion engaged with the recessed portion such that the cylindrical portion is fixed on an outside of the nut base portion, in which in a state where the protrusion portion of the cylindrical portion is engaged with the recessed portion of the nut base portion, the nut base portion is press-fitted and fixed in a circumferential direction and a radial direction of the nut base portion by the cylindrical portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority under 35 U.S.C. § 119 to Japanese Patent Application No. 2024-035561 filed on Mar. 8, 2024 in the Japanese Patent Office, the entire contents of each hereby incorporated by reference.

BACKGROUND OF THE INVENTION

1. Technical Field

[0002] The present invention relates to a nut, a fishing rod reel seat comprising the nut, and a fishing rod comprising the fishing rod reel seat.

2. Description of the Related Art

[0003] In the related art, various devices for fixing a fishing rod attachment leg of a fishing reel to a fishing rod are known. Such a reel fixing device disclosed usually includes a reel seat main body provided with a placing portion on which a fishing rod attachment leg of a fishing reel is placed. A fixed hood for fixing one end of the fishing rod attachment leg is provided at one end of the reel seat main body. A movable hood is attached to the other end of the reel seat main body so as to be movable facing the fixed hood.

[0004] The movable hood usually includes an insertion portion into which the other end of the fishing rod attachment leg is inserted, and an operation knob rotatably attached to the insertion portion. The operation knob includes a cylindrical base portion screwed to a screw portion provided at the other end of the reel seat main body, and a cover member for decoration attached to an outer peripheral surface of the base portion. At that time, usually, the cover member is fixed to the base portion using an adhesive, and the adhesive is applied to both the outer peripheral surface of the base portion and an inner peripheral surface of the cover member.

[0005] As such an operation knob, JP 2020-89310 A discloses a nut of a fishing rod reel seat, the nut being screwed to a male screw portion of a reel seat main body to move a movable hood, the nut including: an inner cylindrical body having an inner peripheral surface provided with a moving female screw portion screwed to the male screw portion of the reel seat main body; and an outer cylindrical body fixed to an outside of the inner cylindrical body, the outer cylindrical body having an outer peripheral surface serving as at least a part of an outer peripheral surface of the nut, in which the outer cylindrical body has a through-hole penetrating in an inward and outward direction, and a fixing portion that fixes the outer peripheral surface of the inner cylindrical body and the inner peripheral surface of the outer cylindrical body at a position farther from the movable hood than the through-hole. In addition, the fixing portion preferably has at least one of an adhesive portion and a screw portion, and in a case where the fixing portion has the adhesive portion, the outer peripheral surface of the inner cylindrical body and the inner peripheral surface of the outer cylindrical body can be easily fixed by adhesion.

SUMMARY OF THE INVENTION

[0006] However, in the nut for moving the movable hood disclosed in JP 2020-89310 A, a fixing portion for fixing the outer peripheral surface of the inner cylindrical body and the inner peripheral surface of the outer cylindrical body is provided, and the outer peripheral surface of the inner cylindrical body and the inner peripheral surface of the outer cylindrical body are fixed by adhesion or a screw fastening structure. Therefore, there is a problem that not only the structure comprising the fixing portion is complicated, but also the process of adhesion or screw fastening is unavoidable, and the process is also complicated in manufacturing.

[0007] The present invention has been made in view of the above circumstances, and an object thereof is to provide a nut capable of reliably performing the fixing between the cylindrical portion and the nut base portion while reducing rattling of the cylindrical portion with respect to the nut base portion in a rotation direction and the axial direction, a fishing rod reel seat comprising the nut, and a fishing rod comprising such a fishing rod reel seat. Other objects of the present invention will become apparent upon reference to the entirety of the present specification.

[0008] A nut according to an embodiment of the present invention is for moving a hood main body of a movable hood, the nut is screwed with a male screw portion of a reel seat main body formed with a reel leg placing portion on which a reel leg is placed, and the nut comprises: a nut base portion which has an outer peripheral surface provided with a recessed portion and an inner peripheral surface provided with a female screw portion for movement that is screwed into the male screw portion of the reel seat main body, and of which an end portion is locked to the hood main body of the movable hood; and a cylindrical portion that has a protrusion portion engaged with the recessed portion such that the cylindrical portion is fixed on an outside of the nut base portion, in which in a state where the protrusion portion of the cylindrical portion is engaged with the recessed portion of the nut base portion, the nut base portion is press-fitted and fixed in a circumferential direction and a radial direction of the nut base portion by the cylindrical portion.

[0009] In the nut according to the embodiment of the present invention, the nut base portion has a plurality of groove portions that extend in a central axis direction of the nut base portion and are formed on an outer peripheral surface of the nut base portion, and the recessed portion is a part of the groove portion.

[0010] In the nut according to the embodiment of the present invention, the groove portion comprises a base portion, an inclined portion, a top portion, and the recessed portion from a side of the groove portion opposite to the hood main body of the movable hood.

[0011] In the nut according to the embodiment of the present invention, the protrusion portion of the cylindrical portion comprises an inclined portion that is formed on the hood main body side of the movable hood and of which a height is increased as a distance from the movable hood is increased.

[0012] In the nut according to the embodiment of the present invention, the protrusion portion of the cylindrical portion comprises the inclined portion and an upper surface portion as viewed in a central axis direction of the cylindrical portion.

[0013] In the nut according to the embodiment of the present invention, the upper surface portion is a flat surface portion extending in the central axis direction of the cylindrical portion.

[0014] In the nut according to the embodiment of the present invention, a length of the protrusion portion of the cylindrical portion in a central axis direction of the cylindrical portion is 3 mm or more.

[0015] In the nut according to the embodiment of the present invention, a length of the flat surface portion in the central axis direction of the cylindrical portion is a length in a range of 0.3 mm to 1.5 mm.

[0016] In the nut according to the embodiment of the present invention, the protrusion portion of the cylindrical portion comprises the upper surface portion and a curved portion extending from both sides of the upper surface portion in the circumferential direction as viewed in the circumferential direction of the cylindrical portion, and a roundness radius (R) of an end portion of the curved portion on a side opposite to the upper surface portion is a size in a range of 0.5 mm to 0.8 mm.

[0017] In the nut according to the embodiment of the present invention, the plurality of groove portions are two or more groove portions.

[0018] In the nut according to the embodiment of the present invention, a depth of the base portion of the groove portion is deeper than a depth of the recessed portion of the groove portion in a range of 0.1 mm to 0.2 mm.

[0019] In the nut according to the embodiment of the present invention, the nut base portion has

hardness lower than hardness of the cylindrical portion.
[0020] In the nut according to the embodiment of the present invention, the protrusion portion of the cylindrical portion has an inclined portion, and an inclination angle of the inclined portion of the groove portion is the same as or different from an inclination angle of the inclined portion of the protrusion portion of the cylindrical portion.
[0021] A fishing rod reel seat according to an embodiment of the present invention comprises any one of the nuts described above.
[0022] A fishing rod according to an embodiment of the present invention comprises: a fishing rod reel seat having the nut according to any one of the nuts described above; and a rod body.
[0023] A nut according to an embodiment of the present invention is for moving a hood main body of a movable hood, the nut is screwed with a male screw portion of a reel seat main body formed with a reel leg placing portion on which a reel leg is placed, and the nut comprises: a nut base portion which has an outer peripheral surface provided with a protrusion portion and an inner peripheral surface provided with a female screw portion for movement that is screwed into the male screw portion of the reel seat main body, and of which an end portion is locked to the hood main body of the movable hood; and a cylindrical portion that has a recessed portion engaged with the protrusion portion such that the cylindrical portion is fixed on an outside of the nut base portion, in which in a state where the recessed portion of the cylindrical portion is engaged with the protrusion portion of the nut base portion, the nut base portion is press-fitted and fixed in a circumferential direction and a radial direction of the nut base portion by the cylindrical portion.
[0024] According to the embodiments, it is possible to provide a nut capable of reliably performing the fixing between the cylindrical portion and the nut base portion while reducing rattling of the cylindrical portion with respect to the nut base portion in a rotation direction and the axial direction, a fishing rod reel seat comprising the nut, and a fishing rod comprising such a fishing rod reel seat.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0025] FIG. **1** is a view illustrating a fishing rod according to an embodiment of the present invention;
[0026] FIG. **2** is a view illustrating a fishing rod reel seat according to an embodiment of the present invention;
[0027] FIG. **3** is a view illustrating a sectional view of a nut according to an embodiment of the present invention;
[0028] FIG. **4** is a view illustrating a nut base portion and a cylindrical portion of a nut according to an embodiment of the present invention;
[0029] FIGS. **5**A and **5**B are views for describing press-fitting and fixing of a nut according to an embodiment of the present invention;
[0030] FIG. **6**A is a perspective view of a nut base portion of a nut according to an embodiment of the present invention, FIG. **6**B is a sectional view of a nut base portion of a nut according to an embodiment of the present invention, and FIG. **6**C is a side view of a nut base portion of a nut according to an embodiment of the present invention;
[0031] FIG. **7**A is a perspective view of a cylindrical portion of a nut according to an embodiment of the present invention, FIG. **7**B is a first sectional view of a cylindrical portion of a nut according to an embodiment of the present invention, FIG. **7**C is an enlarged view of a part (C) of a cross section of a cylindrical portion of a nut according to an embodiment of the present invention, FIG. **7**D is a second sectional view of a cylindrical portion of a nut according to an embodiment of the present invention, and FIG. **7**E is an enlarged view of a part (E) of a cross section of a cylindrical

portion of a nut according to an embodiment of the present invention; and
[0032] FIG. **8** is a view illustrating a nut according to another embodiment of the present invention.

DETAILED DESCRIPTION

[0033] Hereinafter, embodiments of a nut, a reel seat, and a fishing rod according to the present invention will be specifically described with reference to the accompanying drawings. Components that are common in a plurality of drawings are denoted by the same reference signs throughout the plurality of drawings. It should be noted that each drawing is not necessarily drawn to an accurate scale for the sake of convenience in description.

[0034] FIG. **1** is a view illustrating a fishing rod according to an embodiment of the present invention. As illustrated in the drawing, a fishing rod **1** according to an embodiment of the present invention comprises a rod body **2**, a reel **6** attached to the rod body **2** via a reel seat **9**, and a fishing line guide **10** attached to the rod body **2**. In the illustrated embodiment, each of the reel seat **9** and the fishing line guide **10** corresponds to attachment components attached to an outer peripheral surface of the rod body.

[0035] The rod body **2** is formed, for example, by connecting a base rod **3**, a middle rod **5**, a tip rod **7**, and the like to each other. These rod bodies are jointed to each other, for example, as an ordinarily jointed type. The base rod **3**, the middle rod **5**, and the tip rod **7** can be joined to each other by a telescopic type, an inversely jointed type, a socket-and-spigot jointed type, or any other known joining method. The rod body **2** may be constituted by a single rod body.

[0036] Each of the base rod **3**, the middle rod **5**, and the tip rod **7** is constituted by, for example, a tubular body made of a fiber-reinforced resin. This tubular body made of a fiber-reinforced resin is produced by winding a fiber-reinforced resin prepreg (prepreg sheet) obtained by impregnating reinforcing fibers with a matrix resin around a core metal, and heating and curing the prepreg sheet. As the reinforcing fibers contained in the prepreg sheet, for example, carbon fibers, glass fibers, and any other known reinforcing fibers can be used. As the matrix resin contained in the prepreg sheet, a thermosetting resin such as an epoxy resin can be used. After the prepreg sheet is cured, the core metal is removed. An outer surface of the tubular body is appropriately polished. Each rod body may be provided in a solid state.

[0037] In the illustrated embodiment, the base rod **3**, the middle rod **5**, and the tip rod **7** are provided with a plurality of fishing line guides **10** (fishing line guides **10**A to **10**D) for guiding a fishing line fed from the reel **6** attached to the reel seat **9**. More specifically, the base rod **3** is provided with the fishing line guide **10**A, the middle rod **5** is provided with the fishing line guide **10**B, and the tip rod **7** is provided with the fishing line guide **10**C. A top guide **10**D is provided on a distal end of the tip rod **7**, but this is not described in detail.

[0038] Next, the reel seat **9** comprising a reel seat main body **12** and a nut **8** according to the embodiment of the present invention will be described with reference to FIG. **2**. As illustrated, the reel seat **9** according to the embodiment of the present invention comprises the reel seat main body **12** having, along its axial direction, a reel leg placing surface (reel leg placing portion) **12***a* on which a reel leg **6***a* of the fishing reel **6** is placed. The reel seat main body **12** is provided in a cylindrical shape as a whole. In addition, the reel seat main body **12** is provided by an appropriate material such as a synthetic resin (for example, polyamide-based synthetic resin, ABS resin, or the like) or metal (for example, SUS, aluminum, titanium, brass, or the like). The reel seat main body **12** can be configured with, for example, a length of 60 to 160 mm, but is not limited thereto.

[0039] In addition, the reel seat main body **12** comprises a grip portion **12***b* that slightly bulges opposite to the reel leg placing surface **12***a*, and that has a curved outer surface which is made easy to grip by supporting the thenar or the vicinity thereof when gripped with a gripping hand.

[0040] The reel leg placing surface **12***a* of the reel seat main body **12** can be formed flat or substantially flat with a larger curvature than other circumferential portions (for example, the grip portion **12***b*) adjacent to the reel leg placing surface **12***a* of the reel seat main body **12**, and is formed to extend in the axial direction of the reel seat main body **12** illustrated in FIG. **2**. A fixed hood **14** is integrally disposed at one end (rod base side) of the reel seat main body **12**. One end of the reel leg placing surface **12***a* of the reel seat main body **12** is disposed inside the fixed hood **14**.

[0041] The reel seat main body **12** is integrally provided with a cylindrical body in which a guide groove is formed at the other end (rod butt side) and a screw portion (male screw portion) **18**. In the cylindrical body, a cylindrical portion placing surface **13** is formed continuously with the reel leg placing surface **12***a* of the reel seat main body **12** or with a slight step. The cylindrical portion placing surface **13** is also formed continuously with the screw portion (male screw portion) **18**. At the other end (rod butt side), a movable hood **22** is attached to the outside of the cylindrical body and the screw portion **18** so as to be movable in the axial direction. Here, the reel seat **9** may be referred to as comprising the reel seat main body **12** and the movable hood **22**, or the reel seat main body **12**, the movable hood **22**, and the nut **8** that will be described later, but details are omitted. In addition, the movable hood **22** may refer to a hood main body **26** and a reel leg holding portion **28** to be described later, or may be referred to as the movable hood **22** comprising the nut **8**. However, in the present specification, the latter case will be assumed.

[0042] As illustrated in FIG. **2**, in the fixed hood **14** of the reel seat main body **12**, an opening portion **14***a* that receives one end of the reel leg **6***a* is opened toward the movable hood **22**. The opening portion **14***a* of the fixed hood **14** is formed such that the height of an inner surface is gradually decreased toward a front of the reel seat main body **12**. When one end of the reel leg **6***a* is received in the opening portion **14***a* and the one end of the reel leg **6***a* is pressed by the opening portion **14***a*, the reel leg **6***a* is biased (pressed) toward the reel leg placing surface **12***a* disposed in the fixed hood **14**.

[0043] Note that the fixed hood **14** is not limited to being disposed integrally with the reel seat main body **12**, and may be formed of a metal or a hard synthetic resin to have a structure separate from the reel seat main body **12** and firmly fixed to the reel seat main body **12**. In a case of being formed of metal, it is preferable to dispose a resin member at a portion in contact with the reel leg **6***a* of the reel **6** to prevent damage to each member due to the contact between metals.

[0044] The movable hood **22** is formed in a sleeve shape penetrating an axial hole through which the screw portion (male screw portion) **18** of the reel seat main body **12** is inserted, and comprises the hood main body **26**, and the reel leg holding portion **28** formed inside the hood main body **26**. The hood main body **26** is formed of, for example, a synthetic resin. In addition, the nut **8** according to the embodiment of the present invention is rotatably connected to a rear portion (rod butt side) of the hood main body **26** of the movable hood **22**. The nut **8** comprises the nut base portion to be described later and the cylindrical portion to be described later. The nut base portion is formed of, for example, a synthetic resin, a polyamide-based synthetic resin, an ABS resin, or the like, and the cylindrical portion is formed of, for example, a metal such as SUS, aluminum, or brass, but is not limited thereto.

[0045] In the nut **8** according to the embodiment of the present invention, a female screw portion **24** screwed into the screw portion (male screw portion) **18** of the reel seat main body **12** is formed on the inner peripheral side. In addition, a front end portion of the nut **8** (which refers to an end portion of the movable hood **22** on the hood main body **26** side) is connected (locked) to a rear end portion of the hood main body **26** of the movable hood **22** (which refers to an end portion of the nut **8** on the front end portion side) in a state of being relatively rotatable and prevented from falling off. More specifically, the nut **8** can be configured to comprise a locking portion **25** (refer to FIG. **3** to be described later) to be locked to a locked portion **23** (refer to FIG. **3** to be described later) at the rear end of the hood main body **26**, and to be relatively rotatable and prevented from falling off. Note that a connection structure between the nut **8** and the hood main body **26** can be variously considered, and is not limited to a specific aspect.

[0046] The nut **8** according to the embodiment of the present invention is screwed with the screw portion (male screw portion) **18** of the reel seat main body **12** and is rotatably disposed. As described above, since the hood main body **26** is connected to the nut **8**, when the nut **8** is rotated in a forward direction with respect to the screw portion (male screw portion) **18**, the hood main body **26** of the movable hood **22** approaches the fixed hood **14** as the nut **8** approaches the fixed hood **14**. On the other hand, when the nut **8** is rotated in a reverse direction with respect to the screw portion (male screw portion) **18**, the hood main body **26** of the movable hood **22** is moved away from the fixed hood **14** as the nut **8** is moved away from the fixed hood **14**.

[0047] The hood main body **26** moved in the axial direction of the reel seat main body **12** by the rotation of the nut **8** has an opening portion **26***a* that is opened on the front side and the inner peripheral side and is fixed in a state where the reel leg holding portion **28** for receiving the rear end of the reel leg **6***a* of the fishing reel **6** is disposed. The opening portion **26***a* of the hood main body **26** faces the opening portion **14***a* of the fixed hood **14** when the movable hood **22** is attached to the screw portion (male screw portion) **18** of the reel seat main body **12**. Contrary to the opening portion **14***a* of the fixed hood **14**, the opening portion **26***a* of the hood main body **26** is formed such that the height of the inner surface is gradually decreased toward the rear.

[0048] Next, details of the nut **8** according to the embodiment of the present invention will be further described with reference to FIGS. **2**, **3**, **4**, **5**, **6**, and **7**. FIG. **3** is a view illustrating a sectional view of the nut according to the embodiment of the present invention, FIG. **4** is a view illustrating the nut base portion and the cylindrical portion of the nut according to the embodiment of the present invention, and FIGS. **5**A and **5**B are views for describing press-fitting and fixing of the nut according to the embodiment of the present invention. FIG. **6**A is a perspective view of the nut base portion of the nut according to the embodiment of the present invention, FIG. **6**B is a sectional view of the nut base portion of the nut according to the embodiment of the present invention, and FIG. **6**C is a side view of the nut base portion of the nut according to the embodiment of the present invention. FIG. **7**A is a perspective view of the cylindrical portion of the nut according to the embodiment of the present invention, FIG. **7**B is a first sectional view of the cylindrical portion of the nut according to the embodiment of the present invention, FIG. **7**C is an enlarged view of a part (C) of a cross section of the cylindrical portion of the nut according to the embodiment of the present invention, FIG. **7**D is a second sectional view of the cylindrical portion of the nut according to the embodiment of the present invention, and FIG. **7**E is an enlarged view of a part (E) of a cross section of the cylindrical portion of the nut according to an embodiment of the present invention.

[0049] The nut **8** according to the embodiment of the present invention is for moving the hood main body **26** of the movable hood **22**, the nut **8** is screwed with the male screw portion **18** of the reel seat main body **12** formed with a reel leg placing portion **12***a* on which the reel leg **6***a* is placed, and the nut **8** comprises: the nut base portion **15** which has an outer peripheral surface provided with the recessed portion **16** and an inner peripheral surface provided with the female screw portion **24** for movement that is screwed into the male screw portion **18** of the reel seat main body, and of which an end portion **25** is locked to an end portion **23** of the hood main body **26** of the movable hood **22**; and the cylindrical portion **17** that has the protrusion portion **19** engaged with the recessed portion **16** such that the cylindrical portion is fixed on an outside of the nut base portion **15**, in which in a state where the protrusion portion **19** of the cylindrical portion **17** is engaged with the recessed portion **16** of the nut base portion **15**, the nut base portion **15** is press-fitted and fixed to receive a pressure in a circumferential direction (approximately the circumferential direction (direction indicated by arrow A in FIG. **5**B) due to press-fitting) and a radial direction (radial direction (direction indicated by arrow B) due to press-fitting illustrated in FIGS. **5**A and **5**B) of the nut base portion **15** by the cylindrical portion **17**. Here, the dimensional width of the press fitting of the cylindrical portion **17** into the nut base portion **15** is a width in a range of 0 mm to 0.25 mm, and more preferably a width in a range of 0.5 mm to 2.0 mm (note that even in a case where the dimensional width is 0 mm, the contact is made at at least one of the four positions in the circumferential direction because the cross-sectional shape is not a perfect circle). In this manner, as described above, the nut base portion **15** receives the pressure due to the press-fitting of the cylindrical portion **17**, and the cylindrical portion **17** is press-fitted and fixed to the nut base portion **15** such that the nut base portion **15** having hardness relatively smaller (lower) than that of the cylindrical portion **17** is slightly deformed within a range not affecting the structure and function of the nut base portion **15**. In this manner, it is possible to simply and effectively fix the cylindrical portion **17** to the nut base portion **15** while avoiding predetermined or more rattling therebetween only by not using an adhesive or by using an adhesive supplementarily.

[0050] With the nut **8** according to the embodiment of the present invention, it is possible to provide the nut capable of reliably performing the fixing between the cylindrical portion and the nut base portion while reducing rattling of the cylindrical portion with respect to the nut base portion in a rotation direction and the axial direction. More specifically, the nut base portion **15** reliably performs the fixing between the cylindrical portion **17** and the nut base portion **15** while reducing rattling in the rotation direction by applying a pressure in the circumferential direction (substantially circumferential direction (direction indicated by arrow A) due to press-fitting) of the nut base portion **15** by the cylindrical portion **17**, and generates a frictional force between an upper surface portion **19***b* of the cylindrical portion **17** and the recessed portion **16** of the nut base portion **15** by generating a pressing force of the cylindrical portion **17** in the radial direction of the nut base portion **15** by applying a pressure in the radial direction (radial direction (direction indicated by arrow B) due to press-fitting), which makes it possible to reliably perform the fixing between the cylindrical portion **17** and the nut base portion **15** while reducing rattling in the axial direction (central axis X direction of the nut **8**) of the cylindrical portion **17**. Here, since an end portion **15***a* of the nut base portion **15** on the hood main body **26** side of the movable hood **22** is formed to protrude outward in the radial direction, an end portion **17***a* of the cylindrical portion **17** on the hood main body **26** side of the movable hood **22** abuts on the end portion **15***a* of the nut base portion **15** so that the movement of the cylindrical portion **17** toward the hood main body **26** side of the movable hood **22** in the central axis X direction of the nut **8** is restricted. In this manner, rattling of the cylindrical portion **17** in the axial direction (the central axis X direction of the nut **8**) can be further reduced, and fixing between the cylindrical portion **17** and the nut base portion **15** can be more reliably performed.

[0051] Next, the structure of the nut base portion **15** in the nut **8** according to the embodiment of the present invention will be described with reference to FIGS. **6**A to **6**C. As illustrated in FIGS. **6**A to **6**C, in the nut **8** according to the embodiment of the present invention, the nut base portion **15** has one or a plurality of groove portions **20** that extend in the central axis direction of the nut base portion **15** and are formed on the outer peripheral surface of the nut base portion **15**, and the recessed portion **16** is a part of the groove portion **20**. In addition, in the nut **8** according to the embodiment of the present invention, each of the plurality of groove portions **20** comprises a base portion **20***a*, an inclined portion **20***b*, a top portion **20***c*, and the recessed portion **16** when viewed from the side of the groove portion **20** opposite to the hood main body **26** of the movable hood **22**. In addition, in the nut **8** according to the embodiment of the present invention, each of the plurality of groove portions **20** may comprise the base portion **20***a*, the inclined portion **20***b*, the top portion **20***c*, a second inclined portion (not illustrated), and the recessed portion **16** when viewed from the side of the groove portion **20** opposite to the hood main body **26** of the movable hood **22**. In this manner, when the cylindrical portion **17** is attached to the nut base portion **15**, the protrusion portion **19** of the cylindrical portion **17** is guided by the base portion **20***a* of one groove portion **20** of the plurality of groove portions **20** and then guided to ride on the inclined portion **20***b*, and when the protrusion portion **19** exceeds the top portion **20***c* (in the case of having the second inclined portion, when the protrusion portion **19** is guided to descend along the second inclined portion exceeding the top portion **20***c*), the protrusion portion **19** of the cylindrical portion **17** is engaged to be fitted to the recessed portion **16** of the nut base portion **15**, and the cylindrical portion **17** is fixed to the nut base portion **15**.

[0052] Here, one or more groove portions **20** can be provided in the nut base portion **15**. In a case where a plurality of groove portions **20** are provided in the nut base portion **15**, the number of groove portions **20** can be any number of two or more (may be an even number or an odd number). In a case where the nut base portion **15** is provided with a plurality of groove portions **20**, the number of groove portions **20** can be, for example, two or four, but is not limited thereto. In addition, in a case where the nut base portion **15** is provided with a plurality of groove portions **20** (for example, positions at 0°, 90°, 180°, and 270° in the circumferential direction), the cylindrical portion **17** can be provided with a plurality of protrusion portions **19** (for example, positions at 0°, 90°, 180°, and 270° in the circumferential direction) in accordance with the respective positions (for example, positions at 0°, 90°, 180°, and 270° in the circumferential direction) of the plurality of groove portions **20**. In this manner, when the cylindrical portion **17** is attached to the nut base portion **15**, each protrusion portion **19** of the cylindrical portion **17** is guided by the base portion **20***a* of each groove portion **20** of the plurality of groove portions **20** and then guided to ride on the inclined portion **20***b*, and when the protrusion portion **19** exceeds the top portion **20***c* (in the case of having the second inclined portion, when the protrusion portion **19** is guided to descend along the second inclined portion exceeding the top portion **20***c*), each protrusion portion **19** of the cylindrical portion **17** is engaged to be fitted to each recessed portion **16** of the nut base portion **15**, and the cylindrical portion **17** is fixed to the nut base portion **15**. By using the plurality of recessed portions **16** and the plurality of protrusion portions **19**, the cylindrical portion **17** can be more firmly fixed to the nut base portion **15**.

[0053] Next, with reference to FIGS. **7**A to **7**E, the structure of the cylindrical portion **17** in the nut **8** according to the embodiment of the present invention will be described. As illustrated in FIGS. **7**A to **7**E, in the nut **8** according to the embodiment of the present invention, the protrusion portion **19** of the cylindrical portion **17** is formed on the hood main body **26** side of the movable hood **22**, and comprises an inclined portion **19***a* of which the height is increased as the distance from the hood main body **26** of the movable hood **22** is increased. In addition, in the nut **8** according to the embodiment of the present invention, the protrusion portion **19** of the cylindrical portion **17** comprises the above-described inclined portion **19***a* and the upper surface portion **19***b* as viewed in the central axis direction of the cylindrical portion **17** (central axis X direction of the nut **8**).

[0054] In addition, in the nut **8** according to the embodiment of the present invention, the above-described upper surface portion **19***b* is configured to be a flat surface portion extending in the central axis direction of the cylindrical portion **17** (central axis X direction of the nut **8**). In this manner, in a state where the cylindrical portion **17** is press-fitted and fixed to the nut base portion **15**, the upper surface portion **19***b* of the cylindrical portion **17** is formed to press the surface of the nut base portion **15** on the bottom portion side of the recessed portion **16**, and generates a pressing force of the cylindrical portion **17** in the radial direction of the nut base portion **15** by applying a pressure in the radial direction (radial direction (direction indicated by arrow B) due to press-fitting), thereby generating a frictional force between the upper surface portion **19***b* of the cylindrical portion **17** and the recessed portion **16** of the nut base portion **15**. As a result, it is possible to reliably perform the fixing between the cylindrical portion **17** and the nut base portion **15** while reducing rattling of the cylindrical portion **17** in the axial direction (the central axis X direction of the nut **8**).

[0055] In the nut **8** according to the embodiment of the present invention, the length of the protrusion portion **19** of the cylindrical portion **17** in the central axis direction of the cylindrical portion **17** (central axis X direction of the nut **8**) is 3 mm or more. In addition, in the nut **8** according to the embodiment of the present invention, the length of the upper surface portion **19***b* in the central axis direction of the cylindrical portion **17** (central axis X direction of the nut **8**) in a case where the upper surface portion **19***b* is the flat surface portion is a length in a range of 0.3 mm to 1.5 mm.

[0056] In the nut **8** according to the embodiment of the present invention, the protrusion portion **19** of the cylindrical portion **17** comprises the upper surface portion **19***b* and a curved portion **19***c* extending from both sides of the upper surface portion **19***b* in the circumferential direction when viewed in the circumferential direction of the cylindrical portion **17**, and a roundness radius (R) at an end portion **19***d* of the curved portion **19***c* on a side opposite to the upper surface portion **19***b* is a size in a range of 0.5 mm to 0.8 m. In this manner, as illustrated in FIG. **2**, an end portion **16***b* of a side wall **16***a* forming the recessed portion **16** of the nut base portion **15** is pressed against the rounded end portion **19***d* of both curved portions **19***c* of the cylindrical portion **17**, and a pressure substantially in the circumferential direction (direction indicated by arrow A) due to the press-fitting is applied, thereby reducing rattling in the rotational direction.

[0057] In the nut **8** according to the embodiment of the present invention, the depth of the base portion **20***a* of the groove portion **20** (the depth of the nut base portion **15** in the radial direction) is deeper than the depth of the recessed portion **16** of the groove portion **20** (the depth of the nut base portion **15** in the radial direction) in a range of 0.1 mm to 0.2 mm. In this manner, the protrusion portion **19** can be easily guided by the groove portion **20**.

[0058] The nut **8** according to the embodiment of the present invention comprises the inclined portion **19***a* of the protrusion portion **19** of the cylindrical portion **17**, and an inclination angle of the inclined portion **20***b* of the groove portion **20** is configured to be the same as or different from an inclination angle of the inclined portion **19***a* of the protrusion portion **19** of the cylindrical portion **17**. In this manner, the protrusion portion **19** can be easily locked to the recessed portion **16**.

[0059] The fishing rod reel seat **9** according to the embodiment of the present invention comprises any one of the nuts **8** described above. With the fishing rod reel seat **9** comprising the nut **8** according to the embodiment of the present invention, it is possible to provide the nut capable of reliably performing the fixing between the cylindrical portion and the nut base portion while reducing rattling of the cylindrical portion with respect to the nut base portion in a rotation direction and the axial direction. More specifically, the nut base portion **15** reliably performs the fixing between the cylindrical portion and the nut base portion while reducing rattling in the rotation direction by applying a pressure in the circumferential direction (substantially circumferential direction (direction indicated by arrow A) due to press-fitting) of the nut base portion **15** by the cylindrical portion **17**, and generates a frictional force between the upper surface portion **19***b* of the cylindrical portion **17** and the recessed portion **16** of the nut base portion **15** by generating a pressing force of the cylindrical portion in the radial direction of the nut base portion by applying a pressure in the radial direction (radial direction (direction indicated by arrow B) due to press-fitting), which makes it possible to reliably perform the fixing between the cylindrical portion and the nut base portion while reducing rattling in the axial direction.

[0060] The fishing rod **1** according to the embodiment of the present invention comprises the fishing rod reel seat **9** comprising any one of the nuts **8** described above, and the rod body **2**. With the fishing rod **1** comprising the nut **8** according to the embodiment of the present invention, it is possible to provide the nut capable of reliably performing the fixing between the cylindrical portion and the nut base portion while reducing rattling of the cylindrical portion with respect to the nut base portion in a rotation direction and the axial direction. More specifically, the nut base portion **15** reliably performs the fixing between the cylindrical portion and the nut base portion while reducing rattling in the rotation direction by applying a pressure in the circumferential direction (substantially circumferential direction (direction indicated by arrow A) due to press-fitting) of the nut base portion **15** by the cylindrical portion **17**, and generates a frictional force between the upper surface portion **19***b* of the cylindrical portion **17** and the recessed portion **16** of the nut base portion **15** by generating a pressing force of the cylindrical portion in the radial direction of the nut base portion by applying a pressure in the radial direction (radial direction (direction indicated by arrow B) due to press-fitting), which makes it possible to reliably perform the fixing between the cylindrical portion and the nut base portion while reducing rattling in the axial direction.

[0061] The nut according to the embodiment of the present invention is for moving a hood main body of a movable hood, the nut is screwed with a male screw portion of a reel seat main body formed with a reel leg placing portion on which a reel leg is placed, and the nut comprises: a nut base portion which has an outer peripheral surface provided with a protrusion portion and an inner peripheral surface provided with a female screw portion for movement that is screwed into the male screw portion of the reel seat main body, and of which an end portion is locked to the hood main body of the movable hood; and a cylindrical portion that has a recessed portion engaged with the protrusion portion such that the cylindrical portion is fixed on an outside of the nut base portion, in which in a state where the recessed portion of the cylindrical portion is engaged with the protrusion portion of the nut base portion, the nut base portion is press-fitted and fixed in a circumferential direction and a radial direction of the nut base portion by the cylindrical portion. With the nut according to the embodiment of the present invention, it is possible to provide the nut capable of reliably performing the fixing between the cylindrical portion and the nut base portion while reducing rattling of the cylindrical portion with respect to the nut base portion in a rotation direction and the axial direction.

[0062] Next, a nut **80** according to another embodiment of the present invention will be described with reference to FIG. **8**. A basic structure of the nut **80** according to the other embodiment of the present invention is similar to that of the nut **8** according to the embodiment of the present invention described above, and thus description thereof is omitted, but a specific structure of the nut **80** according to the other embodiment of the present invention will be described. As illustrated in the drawing, in the nut **80** according to the other embodiment of the present invention, the hood main body **26** of the movable hood **22** has an end portion provided with the locked portion **23**, and the nut base portion **15** has an end portion provided with the locking portion **25**. However, the locked portion **23** is provided on the outer side of the nut **8** in the radial direction, the locking portion **25** is provided on the inner side of the nut **8** in the radial direction, and the locking portion **25** is locked to the locked portion **23** from the inner side in the radial direction. In addition, in the nut **8** according to the other embodiment of the present invention, the nut base portion **15** comprises a protrusion wall **21** between the recessed portion **16** and the locking portion **25**, and an end portion **27** of the cylindrical portion **17** on the hood main body **26** side of the movable hood **22** is provided to abut against the protrusion wall **21**. In this manner, the movement of the cylindrical portion **17** toward the hood main body **26** side of the movable hood **22** in the central axis X direction of the nut **80** is restricted. In this manner, rattling of the cylindrical portion **17** in the axial direction (the central axis X direction of the nut **80**) can be reduced, and fixing between the cylindrical portion **17** and the nut base portion **15** can be reliably performed.

[0063] Dimensions, materials, and arrangements of the components described in this specification are not limited to those explicitly described in the embodiments, and the components may be modified to have any dimensions, materials, and arrangements that may fall within the scope of the present invention. Components not explicitly described in the present specification can also be added to the described embodiments, or some of the components described in the embodiments can be omitted.

REFERENCE SIGNS LIST

[0064] **1** fishing rod [0065] **2** rod body [0066] **3** base rod [0067] **5** middle rod [0068] **6** reel [0069] **6**_a_ reel leg [0070] **7** tip rod [0071] **8** nut [0072] **9** reel seat [0073] **10** fishing line guide [0074] **12** reel seat main body [0075] **12**_a_ reel leg placing surface [0076] **13** cylindrical portion placing surface [0077] **14** fixed hood [0078] **15** nut base portion [0079] **16** recessed portion [0080] **17** cylindrical portion [0081] **18** screw portion (male screw portion) [0082] **19** protrusion portion [0083] **20** groove portion [0084] **20**_a_ base portion [0085] **20**_b_ inclined portion [0086] **20**_c_ top portion [0087] **21** protrusion wall [0088] **22** movable hood [0089] **23** locked portion [0090] **24** female screw portion [0091] **25** locking portion [0092] **26** hood main body [0093] **27** end portion [0094] **28** reel leg holding portion [0095] **80** nut

Claims

1. A nut for moving a hood main body of a movable hood, the nut being screwed with a male screw portion of a reel seat main body formed with a reel leg placing portion on which a reel leg is placed, the nut comprising: a nut base portion which has an outer peripheral surface provided with a recessed portion and an inner peripheral surface provided with a female screw portion for movement that is screwed into the male screw portion of the reel seat main body, and of which an end portion is locked to the hood main body of the movable hood; and a cylindrical portion that has a protrusion portion engaged with the recessed portion such that the cylindrical portion is fixed on an outside of the nut base portion, wherein in a state where the protrusion portion of the cylindrical portion is engaged with the recessed portion of the nut base portion, the nut base portion is press-fitted and fixed in a circumferential direction and a radial direction of the nut base portion by the cylindrical portion.
2. The nut according to claim 1, wherein the nut base portion has a plurality of groove portions that extend in a central axis direction of the nut base portion and are formed on an outer peripheral surface of the nut base portion, and the recessed portion is a part of the groove portion.
3. The nut according to claim 2, wherein the groove portion comprises a base portion, an inclined portion, a top portion, and the recessed portion from a side of the groove portion opposite to the hood main body of the movable hood.
4. The nut according to claim 1, wherein the protrusion portion of the cylindrical portion comprises an inclined portion that is formed on the hood main body side of the movable hood and of which a height is increased as a distance from the movable hood is increased.
5. The nut according to claim 1, wherein the protrusion portion of the cylindrical portion comprises the inclined portion and an upper surface portion as viewed in a central axis direction of the cylindrical portion.
6. The nut according to claim 5, wherein the upper surface portion is a flat surface portion extending in the central axis direction of the cylindrical portion.
7. The nut according to claim 1, wherein a length of the protrusion portion of the cylindrical portion in a central axis direction of the cylindrical portion is 3 mm or more.
8. The nut according to claim 6, wherein a length of the flat surface portion in the central axis direction of the cylindrical portion is a length in a range of 0.3 mm to 1.5 mm.
9. The nut according to claim 5, wherein the protrusion portion of the cylindrical portion comprises the upper surface portion and a curved portion extending from both sides of the upper surface portion in the circumferential direction as viewed in the circumferential direction of the cylindrical portion, and a roundness radius (R) of an end portion of the curved portion on a side opposite to the upper surface portion is a size in a range of 0.5 mm to 0.8 m.
10. The nut according to claim 2, wherein the plurality of groove portions are two or more groove portions.
11. The nut according to claim 3, wherein a depth of the base portion of the groove portion is deeper than a depth of the recessed portion of the groove portion in a range of 0.1 mm to 0.2 mm.
12. The nut according to claim 1, wherein the nut base portion has hardness lower than hardness of the cylindrical portion.
13. The nut according to claim 3, wherein the protrusion portion of the cylindrical portion has an inclined portion, and an inclination angle of the inclined portion of the groove portion is the same as or different from an inclination angle of the inclined portion of the protrusion portion of the cylindrical portion.
14. A fishing rod reel seat comprising the nut according to claim 1.
15. A fishing rod comprising: a fishing rod reel seat comprising the nut according to claim 1; and a rod body.

16. A nut for moving a hood main body of a movable hood, the nut being screwed with a male screw portion of a reel seat main body formed with a reel leg placing portion on which a reel leg is placed, the nut comprising: a nut base portion which has an outer peripheral surface provided with a protrusion portion and an inner peripheral surface provided with a female screw portion for movement that is screwed into the male screw portion of the reel seat main body, and of which an end portion is locked to the hood main body of the movable hood; and a cylindrical portion that has a recessed portion engaged with the protrusion portion such that the cylindrical portion is fixed on an outside of the nut base portion, wherein in a state where the recessed portion of the cylindrical portion is engaged with the protrusion portion of the nut base portion, the nut base portion is press-fitted and fixed in a circumferential direction and a radial direction of the nut base portion by the cylindrical portion.